

Ad Hoc Human–AI Orchestration via Skill Inference

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Hybrid teams need to know who is good at what

- Hybrid workforces: human experts and AI agents on the same team.
- They work only if an orchestrator gives each task to whoever is best.
- New hires and new models arrive with unknown strengths. Résumés and model cards don't say who wins the next task.

Cold start: route work across entities you've barely observed.

The question

How can an orchestrator learn the skill profiles of both humans and agents from only a few noisy observations, and use them to drive ad hoc human–AI orchestration?

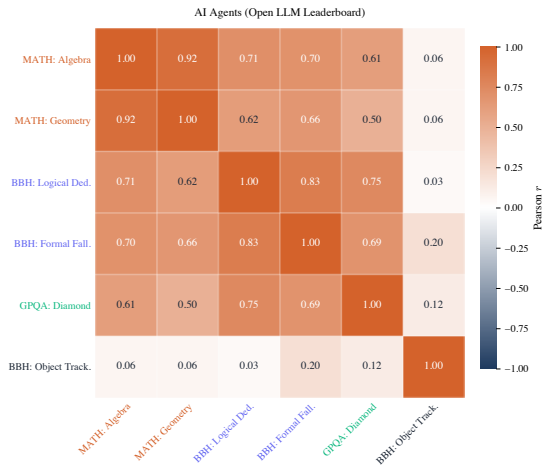
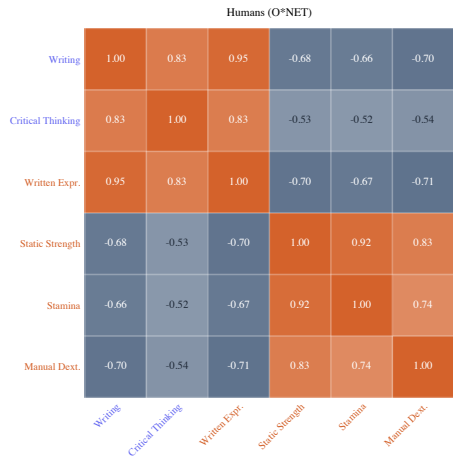
RQ1 does skill transfer beat per-skill estimation?

RQ3 are the posteriors calibrated?

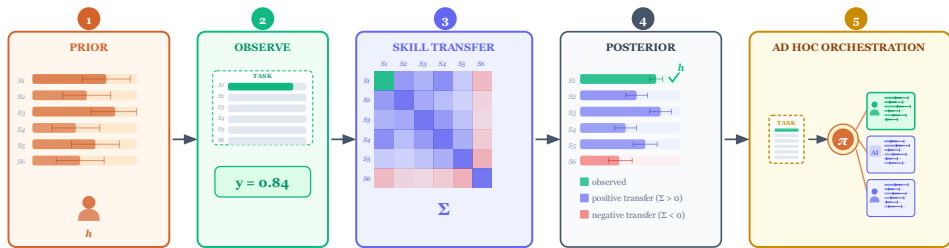
RQ2 does signal granularity matter?

RQ4 can they drive online routing?

The insight: skills are not independent



The skill inference pipeline



One observation updates the whole profile; posteriors then drive routing.

Each entity is a Gaussian belief over a skill taxonomy

- Entity h : a Gaussian $\mathcal{N}(\boldsymbol{\mu}_h, \boldsymbol{\Sigma}_h)$.
- $\boldsymbol{\mu}_h$, one value per skill.
- $\boldsymbol{\Sigma}_h$, the shared population covariance.

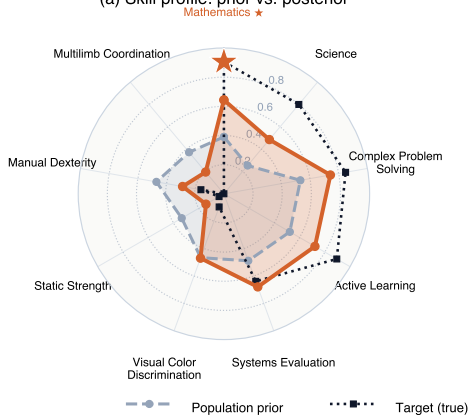
$$\boldsymbol{\Sigma} = \frac{1}{N-1} (\mathbf{R} - \bar{\mathbf{R}})^\top (\mathbf{R} - \bar{\mathbf{R}})$$

$\mathbf{R}$: the population skill matrix
(N entities $\times$ K skills)

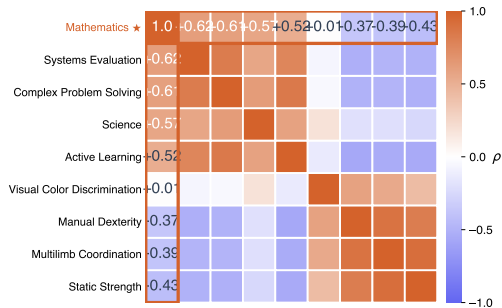
Humans and agents share one skill space.

One observation updates every skill

(a) Skill profile: prior vs. posterior



(b) Skill correlations carrying the update



The Kalman update rule

$$\mathbf{K} = \Sigma_{:,j} / (\Sigma_{jj} + \sigma_{\text{obs}}^2)$$

Kalman gain

$$\boldsymbol{\mu} \leftarrow \boldsymbol{\mu} + \mathbf{K} (y_j - \mu_j)$$

spread to all skills

$$\Sigma \leftarrow \Sigma - \mathbf{K} \Sigma_{j,:}$$

shrink uncertainty

Each observation updates the whole profile through Σ . Cost $O(K^2)$.

When the population has clusters: mixture of Gaussians

- One covariance assumes everyone correlates alike, humans and AI don't.
- GMM- Σ : a mixture prior $\sum_m \pi_m \mathcal{N}(\mu_m, \Sigma_m)$ (EM-fit, BIC).
- Each entity leans on its own cluster, agents on AI, people on human covariance.

Same update, fit per cluster. Helps most on mixed populations.

From posteriors to routing: Thompson sampling

For a task needing skill c_t , draw from each entity's posterior and pick the best:

$$\tilde{s}_{a,c_t} \sim \mathcal{N}(\mu_{a,c_t}, \Sigma_{a,c_t c_t})$$

$$a_t = \arg \max_a \tilde{s}_{a,c_t}$$

- High estimate $\Rightarrow$ exploit.
- High uncertainty $\Rightarrow$ explore.
- As observations accrue, the policy converges to greedy.

Learn skills and assign work at the same time.

Evaluation: five populations, one algorithm

Recovery (RQ1–3)

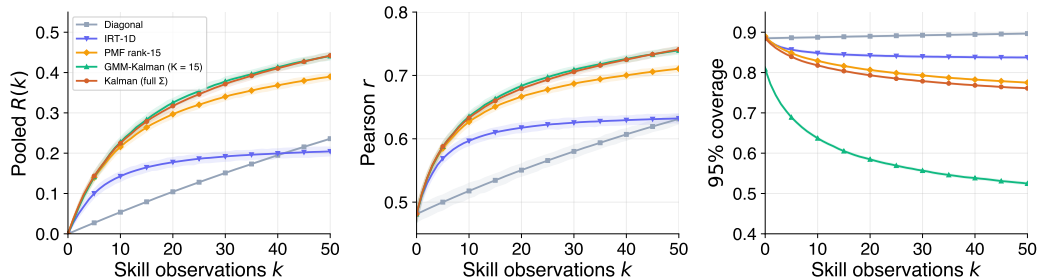
O*NET	synthetic humans, 120 skills	synthetic
PIAAC	2,514 real adults, 9 skills	human
Open LLM Leaderboard	2,313 models, 39 benchmarks	AI

Online orchestration (RQ4)

Codeforces	618 programmers	human
AtCoder + LiveCodeBench	60 humans + 60 models	hybrid

The same method on humans, AI, and mixed teams.

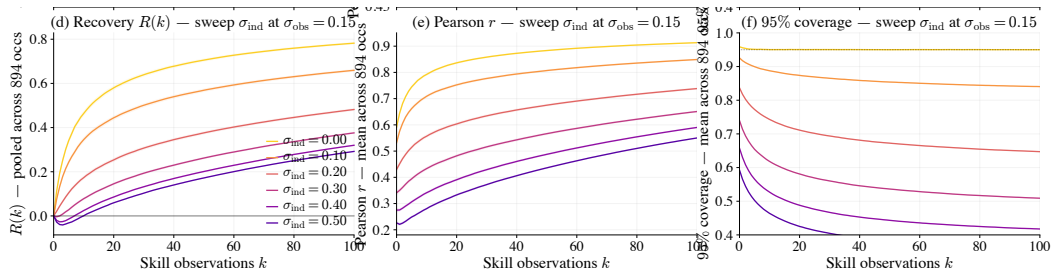
Skill transfer recovers profiles faster (O*NET)



Covariance methods (Σ , GMM) recover fastest and stay calibrated.

Individual noise breaks calibration (O*NET ablation)

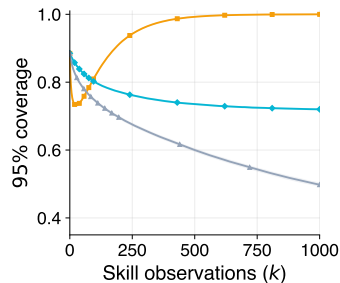
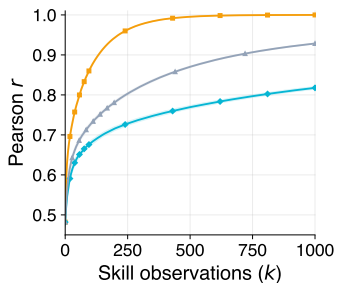
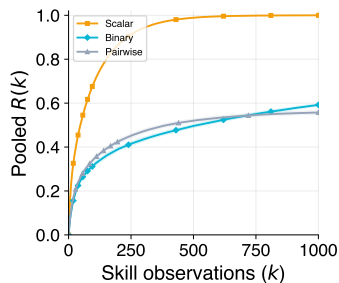
sweeping individual noise σ_{ind} at fixed observation noise, recovery, shape, 95% coverage



Entities far from the prior slow recovery and lose calibration; observation noise alone has a much milder effect.

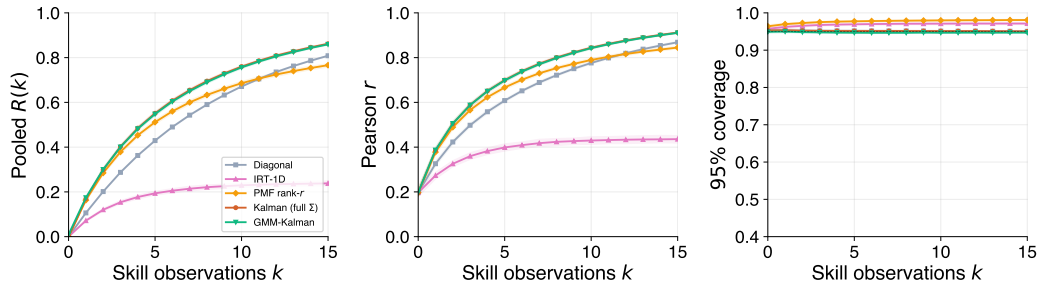
Richer signals recover faster

Scalar: score $\in [0, 1]$ • Binary: pass/fail • Pairwise: $A > B$ all feed the same filter



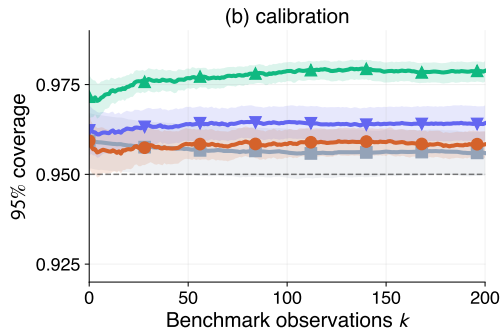
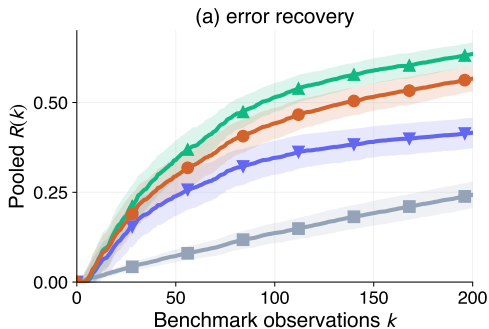
Scalar wins on error and calibration; binary slower; pairwise gets shape, not magnitude.

Real humans: PIAAC

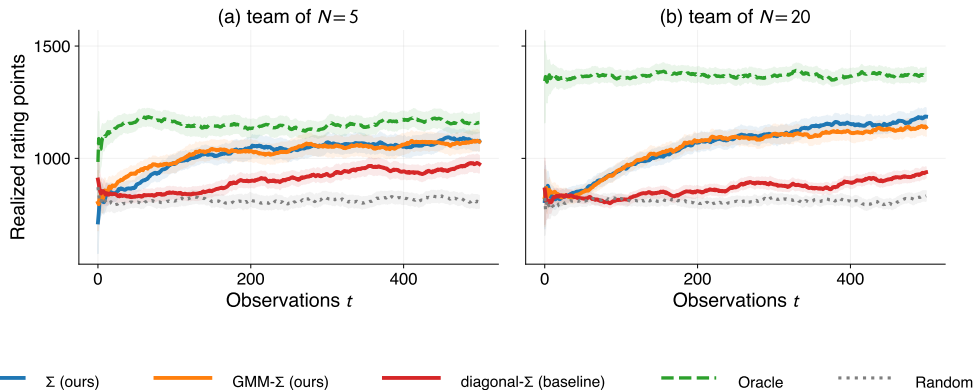


2,514 real adults: Σ beats independent estimation, at 95% coverage.

Real agents: Open LLM Leaderboard

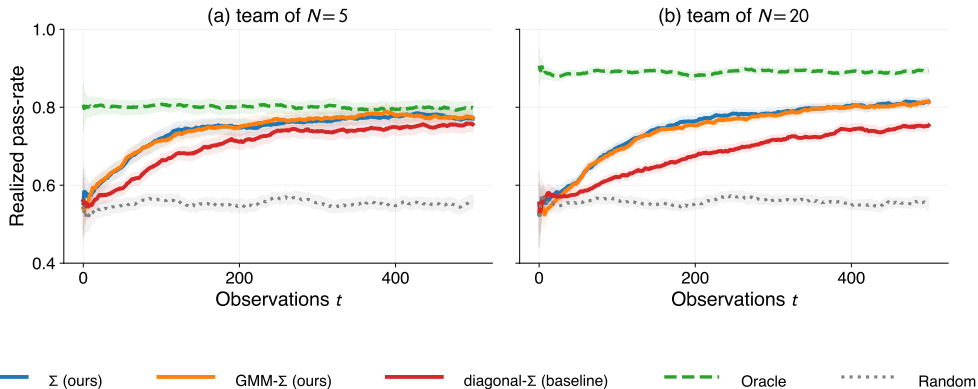


Online orchestration: human teams (Codeforces)



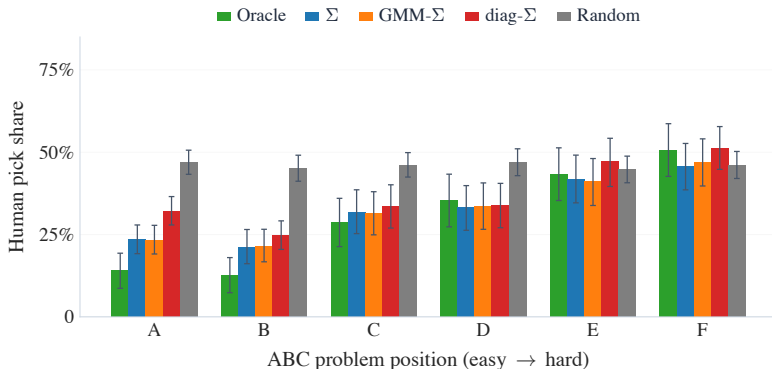
618 programmers: Σ /GMM- Σ beat independent; the edge grows with team size.

Online orchestration: hybrid teams (AtCoder + LLMs)



60 humans + 60 LLMs: the same routing learns the hybrid policy.

It discovers human–AI complementarity



The split emerges on its own: easy $\rightarrow$ AI, hard $\rightarrow$ humans.

The skillinfer library

```
import skillinfer

pop = skillinfer.datasets.onet()
profile = pop.profile()
profile.observe("Skill:Programming", 0.70)
profile.observe("Skill:Critical Thinking", 0.65)
profile.predict()    # 120 skills, 95% CIs
```

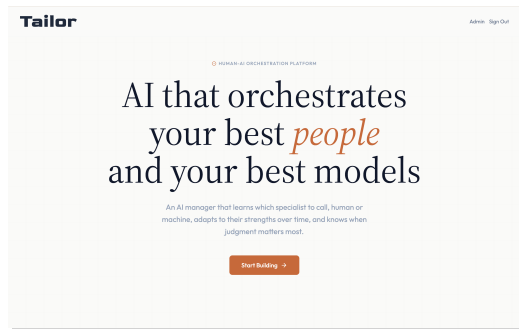
returns every skill, with 95% CIs

	feature	mean	std	ci_lower	ci_upper
0	Skill:Active Learning	0.721046	0.117250	0.491241	0.950851
1	Skill:Active Listening	0.614567	0.095642	0.427113	0.802022
2	Skill:Complex Problem Solving	0.788529	0.094126	0.604045	0.973013
3	Skill:Coordination	0.479024	0.138980	0.206627	0.751420
4	Skill:Critical Thinking	0.650604	0.009099	0.632770	0.668439
..	...	...	...	...	...

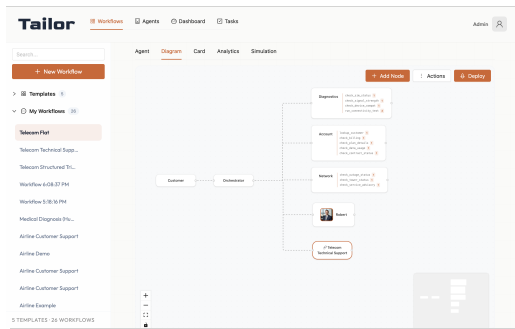
[120 rows x 5 columns]

`pip install skillinfer` • docs: kostadindev.github.io/skillinfer

Tailor: build the orchestration visually



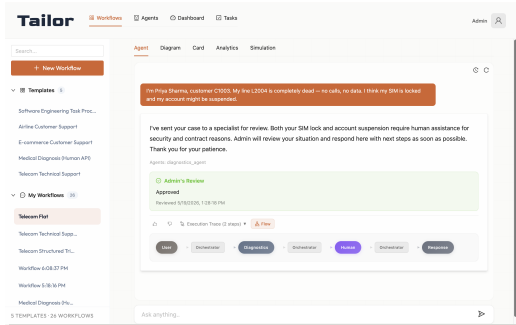
Landing page, one platform for hybrid teams



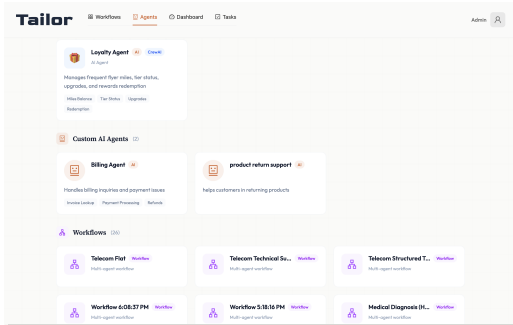
Workflow editor, wire humans, agents, and orchestrator

Where the skill profiles could live: a system, not a claim of this thesis.

Tailor: one workspace for humans and agents



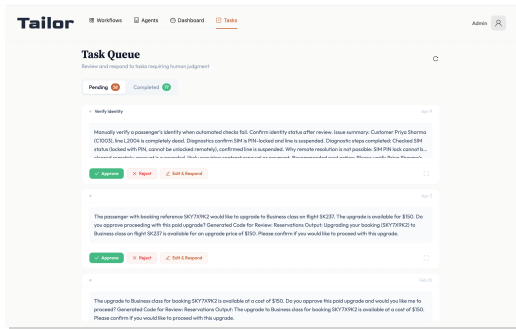
Chat with a hybrid team, human review inline



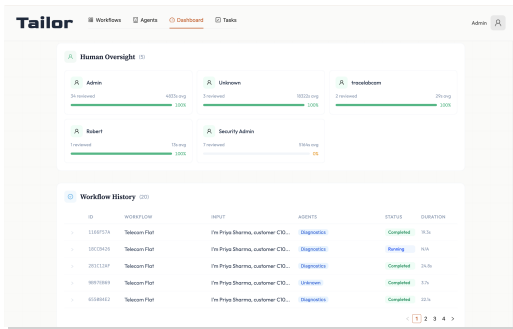
Agent library, skills listed per agent

Humans and AI answer side by side, each as a skill profile.

Tailor: human oversight, built in



Human task queue, approve / reject / edit



Oversight dashboard, who reviewed what

Escalations queue to people; every review is logged.

Calibrated, interpretable profiles enable oversight

- The model reports its uncertainty. Escalate to a human where confidence is low.
- Not a black box. Each decision traces to per-skill posteriors (GDPR Art. 22, EU AI Act oversight).
- Fairness. Priors inherit population bias; monitor by demographic and refit often.

Calibrated, explainable profiles make oversight precise.

Takeaways

- Skills correlate, so a few observations recover the full profile.
- One algorithm for humans, AI, and hybrid teams.
- Beats independent estimation on recovery and on routing.
- Finds human–AI complementarity without being told.
- Posteriors stay calibrated when the prior is fit on individuals.

Observe a few skills, infer the rest, and route work accordingly.

Thank you



SCAN FOR DOCS & CODE

```
pip install skillinfer
```

kostadindev.github.io/skillinfer

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